

# Network Security Monitoring & Threat Detection using Zeek Integrated with Wazuh

## Project Overview

Modern SOC environments require **deep visibility into network traffic** to detect threats such as reconnaissance, data exfiltration, malicious DNS activity, and SSL/TLS abuse. Traditional endpoint logs alone are not sufficient to understand *what is happening on the wire*.

In this project, I implemented **Zeek (formerly Bro)** as a **network traffic analysis engine** and integrated it with **Wazuh SIEM** to provide **real-time network security monitoring, alerting, and visualization**.

This setup allows SOC analysts to:

- See **who is talking to whom** on the network
- Detect **DNS abuse, port scans, rejected connections**
- Identify **SSL/TLS certificate anomalies**
- Correlate raw network activity with **SIEM alerts**
- Monitor everything visually using **custom Wazuh dashboards**



## Why Zeek + Wazuh Integration?

### ◆ What is Zeek?

Zeek is a powerful **network security monitoring and traffic analysis tool**. Instead of blocking traffic, Zeek:

- Passively inspects network packets
- Generates **high-fidelity structured logs**
- Provides deep protocol-level visibility (DNS, SSL, HTTP, TCP, etc.)

## ◆ Why Integrate Zeek with Wazuh?

Wazuh is a SIEM/XDR platform that excels at:

- Log aggregation
- Correlation
- Alerting
- Visualization

By integrating Zeek with Wazuh:

- Zeek provides **network intelligence**
- Wazuh provides **SIEM correlation & dashboards**
- Together, they create a **powerful SOC-ready network monitoring solution**

## Tools & Technologies Used

- **Zeek** – Network traffic analysis
- **Wazuh Manager & Agent** – SIEM, alerting, dashboards
- **Ubuntu Linux** – Zeek endpoint
- **Kali Linux** – Attacker simulation
- **OpenSearch Dashboards (Wazuh UI)** – Visualization
- **DNS / SSL / TCP traffic simulation tools**

## Architecture Overview

### Traffic Flow Architecture:

1. Network traffic is observed by **Zeek**
2. Zeek generates **JSON-formatted logs**
3. Wazuh Agent monitors Zeek log files
4. Logs are sent to **Wazuh Manager**
5. Custom **decoders & rules** parse Zeek logs
6. Alerts are generated and indexed
7. Dashboards visualize network activity for SOC analysts

## Implementation Steps

### Step 1: Adding Zeek Repository & GPG Key

Added the official Zeek repository and imported the GPG key to ensure:

- Secure package installation
- Access to latest Zeek versions

**Purpose:**

Ensures trusted and up-to-date Zeek installation via APT.

```
root@thehive:/home/ubuntu# echo 'deb http://download.opensuse.org/repositories/security:/zeek/xUbuntu_24.04/ /' | sudo tee /etc/apt/sources.list.d/security:zeek.list
deb http://download.opensuse.org/repositories/security:/zeek/xUbuntu_24.04/ /
root@thehive:/home/ubuntu# curl -fsSL https://download.opensuse.org/repositories/security:/zeek/xUbuntu_24.04/Release.key | \
gpg --dearmor | sudo tee /etc/apt/trusted.gpg.d/security_zeek.gpg > /dev/null
```

**Step 2: Installing Zeek**

Installed Zeek

**Purpose:**

Deploys Zeek binaries and required dependencies on the endpoint.

```
root@thehive:/home/ubuntu# apt update -y
apt install zeek -y
```

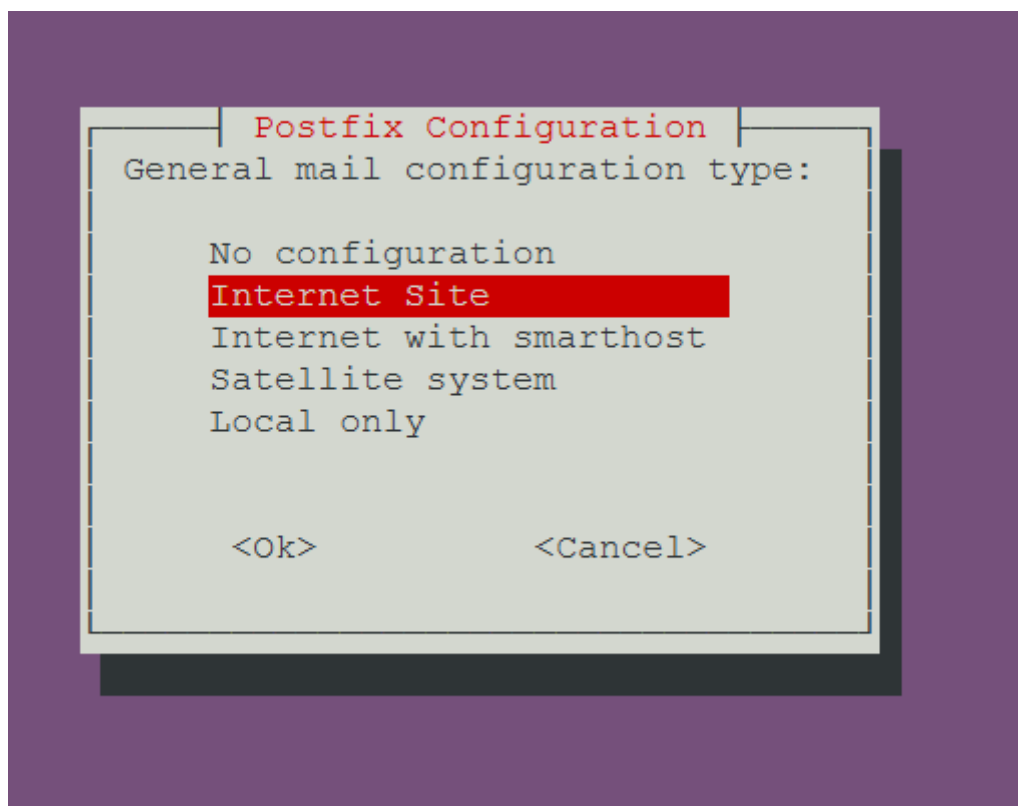
**Step 3: Postfix Configuration & Hostname Setup**

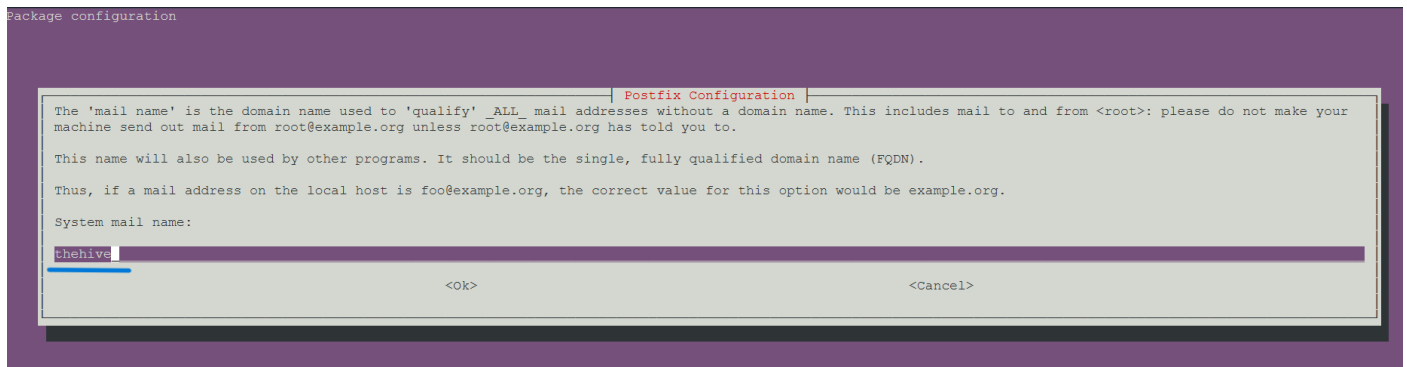
During installation, Postfix mail configuration was prompted:

- Selected **Internet Site**
- Configured system hostname

**Purpose:**

Ensures system services function correctly and avoids installation issues.





#### Step 4: Exporting Zeek Binary Path & Verification

Added Zeek to PATH using .bashrc and verified installation:

```
zeek --version
```

##### Purpose:

Allows global execution of Zeek commands and confirms successful installation.

```
root@thehive:/home/ubuntu# echo "export PATH=$PATH:/opt/zeek/bin" >> ~/.bashrc
source ~/.bashrc
root@thehive:/home/ubuntu# zeek --version
zeek version 8.1.1
```

#### Step 5: Configuring Zeek Node Settings

Edited node.cfg to define:

- Standalone deployment
- Network interface to monitor

##### Purpose:

Tells Zeek **which interface to sniff traffic from**.

```
GNU nano 7.2 /opt/zeek/etc/node.cfg *
# Example ZeekControl node configuration.
#
# This example has a standalone node ready to go except for possibly changing
# the sniffing interface.
#
# This is a complete standalone configuration. Most likely you will
# only need to change the interface.
[zeek]
type=standalone
host=localhost
interface=enX0
```

#### Step 6: Configuring Network Scope

Updated networks.cfg with internal/private network ranges.

##### Purpose:

Helps Zeek differentiate **internal vs external traffic**, improving detection accuracy.

```

GNU nano 7.2 /opt/zeek/etc/networks.cfg *
# List of local networks in CIDR notation, optionally followed by a descriptive
# tag. Private address space defined by Zeek's Site::private_address_space set
# (see scripts/base/utls/site.zeek) is automatically considered local. You can
# disable this auto-inclusion by setting zeekctl's PrivateAddressSpaceIsLocal
# option to 0.
#
# Examples of valid prefixes:
#
# 1.2.3.0/24      Admin network
# 2607:f140::/32  Student network
10.0.1.0/24      AWS-Internal-Network

```

## Step 7: Enabling JSON Log Output

Modified local.zeek to enable:

@load policy/tuning/json-logs.zeek

### Purpose:

- Outputs Zeek logs in **JSON format**
- Required for Wazuh parsing and decoding

```

GNU nano 7.2 /opt/zeek/share/zeek/site/local.zeek *
# Uncomment the following line to enable logging of Community ID hashes in
# the conn.log file.
# @load policy/protocols/conn/community-id-logging

# Uncomment the following line to enable logging of connection VLANs. Enabling
# this adds two VLAN fields to the conn.log file.
# @load policy/protocols/conn/vlan-logging

# Uncomment the following line to enable logging of link-layer addresses. Enabling
# this adds the link-layer address for each connection endpoint to the conn.log file.
# @load policy/protocols/conn/mac-logging

# Uncomment this to source zkg's package state
# @load packages
@load policy/tuning/json-logs.zeek

```

## Step 8: Verifying Zeek Configuration

Checked Zeek configuration:

zeekctl check

### Purpose:

Ensures there are no syntax or configuration errors before deployment.

```

(venv) root@thehive:/home/ubuntu# zeekctl check
Hint: Run the zeekctl "deploy" command to get started.
zeek scripts are ok.

```

## Step 9: Deploying & Starting Zeek

Started Zeek services:

zeekctl deploy

### Purpose:

Activates traffic monitoring and log generation.

```
(venv) root@thehive:/home/ubuntu# zeekctl deploy
checking configurations ...
installing ...
creating policy directories ...
installing site policies ...
generating standalone-layout.zeek ...
generating local-networks.zeek ...
generating zeekctl-config.zeek ...
generating zeekctl-config.sh ...
stopping ...
stopping zeek ...
starting ...
starting zeek ...
```

## Step 10: Configuring Wazuh Agent Log Monitoring

Configured Wazuh agent to monitor:

/opt/zeek/logs/current/\*.log

### Purpose:

Allows Wazuh to ingest Zeek network logs in real time.

```
<localfile>
  <log_format>json</log_format>
  <location>/opt/zeek/logs/current/*.log</location>
</localfile>
```

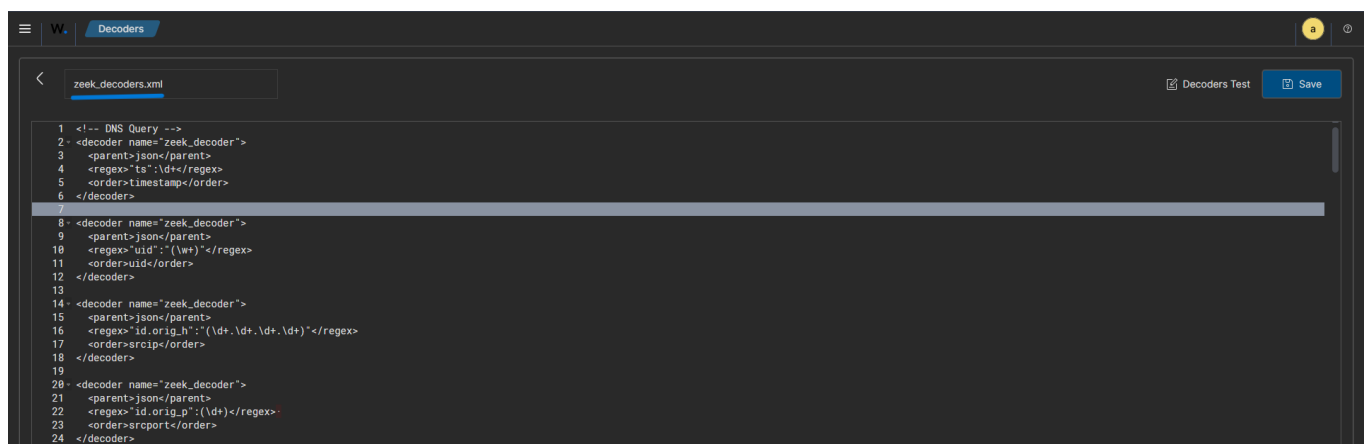
## Step 11: Creating Wazuh Decoders for Zeek Logs

Custom decoders were added to:

- Parse Zeek JSON fields
- Extract DNS, IP, port, protocol, SSL fields

### Purpose:

Transforms raw Zeek logs into structured, searchable Wazuh fields.



```
Wazuh Decoders
zeek_decoders.xml
Decoders Test Save

1 <!-- DNS Query -->
2 <decoder name="zeek_decoder">
3   <parent>json</parent>
4   <regex>"ts":\d+</regex>
5   <order>timestamp</order>
6 </decoder>
7
8 <decoder name="zeek_decoder">
9   <parent>json</parent>
10  <regex>"uid":\w+</regex>
11  <order>uid</order>
12 </decoder>
13
14 <decoder name="zeek_decoder">
15   <parent>json</parent>
16   <regex>"id.orig_h":(\d+\.\d+\.\d+\.\d+)</regex>
17   <order>srcip</order>
18 </decoder>
19
20 <decoder name="zeek_decoder">
21   <parent>json</parent>
22   <regex>"id.orig_p":(\d+)</regex>
23   <order>srcport</order>
24 </decoder>
25
```

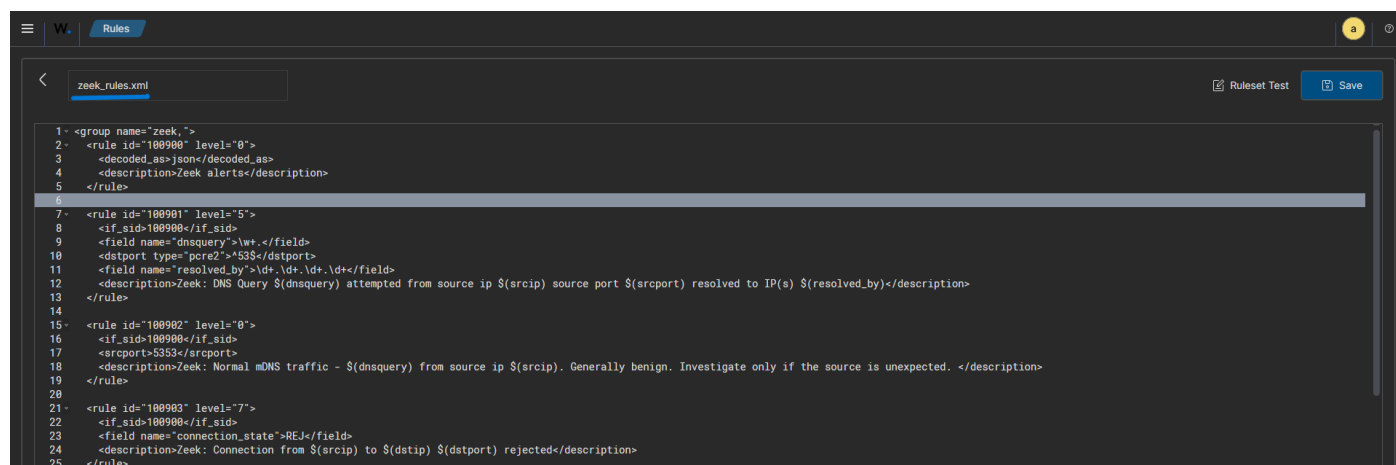
## Step 12: Creating Wazuh Rules for Network Alerts

Custom rules created to detect:

- DNS queries
- Rejected connections
- Port scanning behavior
- SSL/TLS certificate issues

### Purpose:

Assigns **alert severity and descriptions** for SOC analysts proper visibility.



The screenshot shows the Wazuh Rules configuration interface. The top bar includes a menu icon, the Wazuh logo, and a 'Rules' tab. Below the tab, there's a search bar containing 'zeek\_rules.xml' and buttons for 'Ruleset Test' and 'Save'. The main area displays the XML content of the zeek\_rules.xml file, which defines several rules for detecting network alerts. The rules are numbered 1 through 25. Rule 1 is a group named 'zeek'. Rule 2 is a rule with id '100900' and level '0', which is a decoded\_as rule. Rule 3 is a rule with id '100901' and level '5', which is a rule that detects DNS queries. Rule 4 is a rule with id '100902' and level '0', which is a rule that detects DNS queries. Rule 5 is a rule with id '100903' and level '7', which is a rule that detects rejected connections. The XML content is as follows:

```
1- <group name="zeek">
2-   <rule id="100900" level="0">
3-     <decoded_as>json</decoded_as>
4-     <description>Zeek alerts</description>
5-   </rule>
6-
7-   <rule id="100901" level="5">
8-     <if_sid>100900</if_sid>
9-     <field name="dnsquery">\wt.</field>
10-    <dstport type="pore2">^53$</dstport>
11-    <field name="resolved_by">\d+.\d+.\d+.\d+</field>
12-    <description>Zeek: DNS Query $(dnsquery) attempted from source ip $(srcip) source port $(srcport) resolved to IP(s) $(resolved_by)</description>
13-  </rule>
14-
15-   <rule id="100902" level="0">
16-     <if_sid>100900</if_sid>
17-     <srcport>5353</srcport>
18-     <description>Zeek: Normal mDNS traffic - $(dnsquery) from source ip $(srcip). Generally benign. Investigate only if the source is unexpected. </description>
19-   </rule>
20-
21-   <rule id="100903" level="7">
22-     <if_sid>100900</if_sid>
23-     <field name="connection_state">REJ</field>
24-     <description>Zeek: Connection from $(srcip) to $(dstip) $(dstport) rejected</description>
25-   </rule>
```

## Detection & Testing Scenarios

### Step 13: DNS Query Activity Test

Simulated DNS queries using dig.

### Purpose:

Validates DNS logging, parsing, and alert generation.

```
(venv) root@thehive:/home/ubuntu# dig wazuh.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> wazuh.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 59772
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;wazuh.com.                IN      A

;; ANSWER SECTION:
wazuh.com.                53      IN      A      3.162.112.79
wazuh.com.                53      IN      A      3.162.112.12
wazuh.com.                53      IN      A      3.162.112.128
wazuh.com.                53      IN      A      3.162.112.37

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sun Feb 08 01:15:27 IST 2026
;; MSG SIZE rcvd: 102

(venv) root@thehive:/home/ubuntu# dig virustotal.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> virustotal.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 50455
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;virustotal.com.          IN      A

;; ANSWER SECTION:
virustotal.com.          291     IN      A      216.239.38.21
virustotal.com.          291     IN      A      216.239.36.21
virustotal.com.          291     IN      A      216.239.34.21
virustotal.com.          291     IN      A      216.239.32.21
```

## Step 14: DNS Alerts in Wazuh

Observed alerts showing:

- Queried domain
- Source IP
- Resolution results

### Purpose:

Confirms DNS visibility inside SIEM.



## Step 15: Reconnaissance & Port Scan Test

From Kali Linux, executed port scan:

```
nc -zv <target-ip> 5555-5559
```

### Purpose:

Simulates attacker reconnaissance and scanning behavior.

```
(kali@kali)-[~]
$ for port in {5555..5559}; do nc -zv [redacted] $port || true; done
nc: connect to 3.162.112.128: port 5555 (tcp) failed: Connection refused
nc: connect to 3.162.112.128: port 5556 (tcp) failed: Connection refused
nc: connect to 3.162.112.128: port 5557 (tcp) failed: Connection refused
nc: connect to 3.162.112.128: port 5558 (tcp) failed: Connection refused
nc: connect to 3.162.112.128: port 5559 (tcp) failed: Connection refused
```

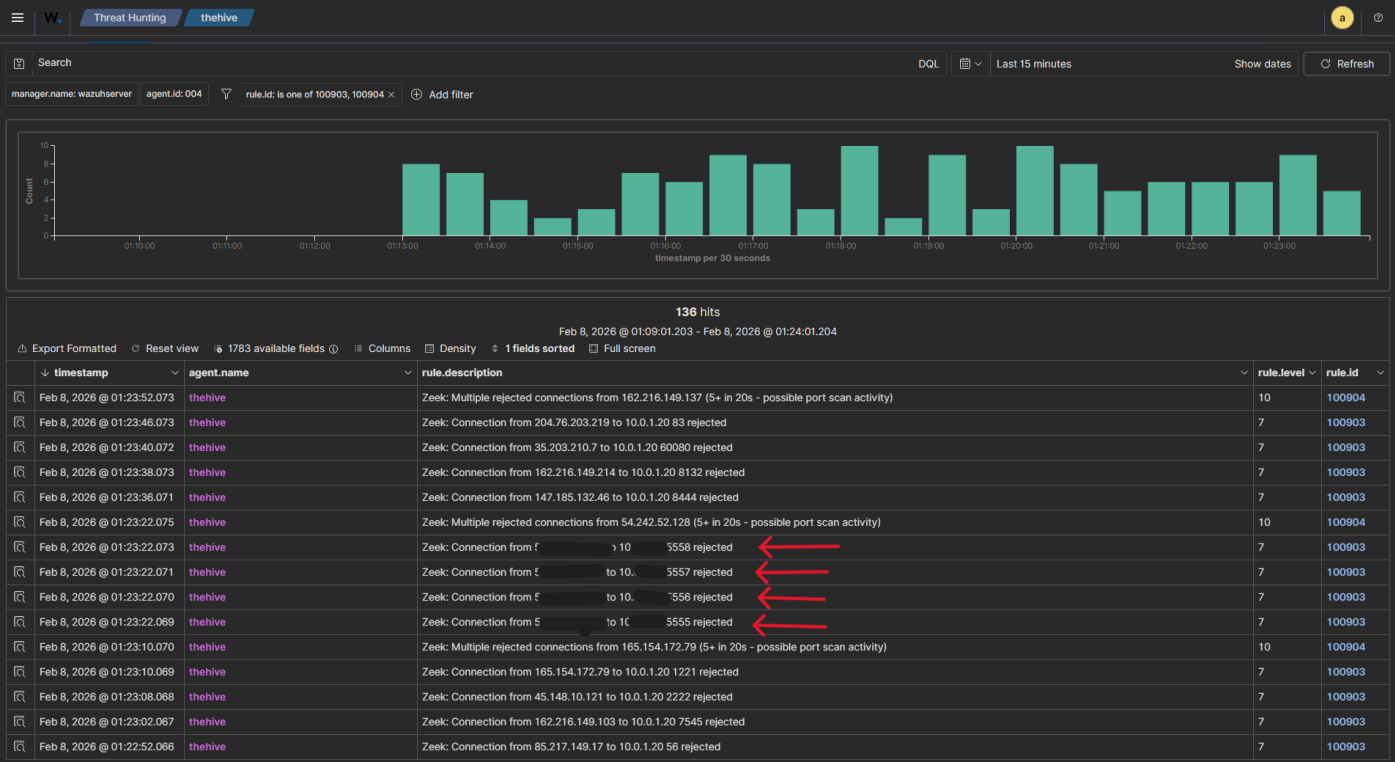
## Step 16: Port Scan Detection in Wazuh

Alerts generated showing:

- Source attacker IP
- Target endpoint
- Rejected connections

### Purpose:

Demonstrates detection of scanning attempts.



Step 17: SSL/TLS Certificate Anomaly Test

Connected to servers with:

- Self-signed certificates
- Invalid SSL configurations

Purpose:

Simulates MITM or suspicious encrypted traffic.

```
(venv) root@thehive:/home/ubuntu# curl -k https://self-signed.badssl.com
<!DOCTYPE html>
<html>
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<link rel="shortcut icon" href="/icons/favicon-red.ico"/>
<link rel="apple-touch-icon" href="/icons/icon-red.png"/>
<title>self-signed.badssl.com</title>
<link rel="stylesheet" href="/style.css">
<style>body { background: red; }</style>
</head>
<body>
<div id="content">
<h1 style="font-size: 12vw;">
self-signed.<br>badssl.com
</h1>
</div>
</body>
</html>
(venv) root@thehive:/home/ubuntu# curl -k https://revoked.grc.com
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en">
<head><title>GRC's Security Certificate Revocation Awareness Test</title>
<meta name="keywords" content="Security Certificate Revocation Awareness Test, GRC, Gibson Research Corporation, Steve Gibson" />
<meta name="description" content="Security Certificate Revocation Awareness Test" />
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
<meta http-equiv="Content-Style-Type" content="text/css" />
<link rel="meta" href="/revoked/labels.rdf" type="application/rdf+xml" title="ICRA labels" />
<link rel="stylesheet" media="all" type="text/css" href="https://www.grc.com/grc.css" />
</head><body style="background:#fff000"><a name="top"></a><center>

<div style="background:#fff; margin:30px; border-radius:30px; padding:30px;"><div style="color:#b00000; font-size:24pt; font-family:Arial, Helvetica, sans-serif"><b>Security Certificat
e</b><br /></Revocation Awareness Test</div><div style="font-size:15pt; margin:0.5em;">If you can see this (and apparently you can), you<br />are using a <b>revocation UNaware</b> web bro
wser!</div>

<div class="pagecontainer" style="margin-bottom:1em;">

<center><div class="attention box" style="border:2px solid black; width:35em;"><center><table><tr><td style="font-size:11pt; color:#770000;"><b>The SSL/TLS security certificate for this sp
cial website has been <em>deliberately revoked</em>. Since you <em>are</em> seeing this page, we know that this web browser is allowing a site with a <em>known invalid certi
ficate</em> to display its pages. This is likely not the behavior you would choose.</td></tr></table></center></div></center>

<p><b>Here's what we know &#211; as of August 21st, 2024:</b></p><ul class="tightlist">
```

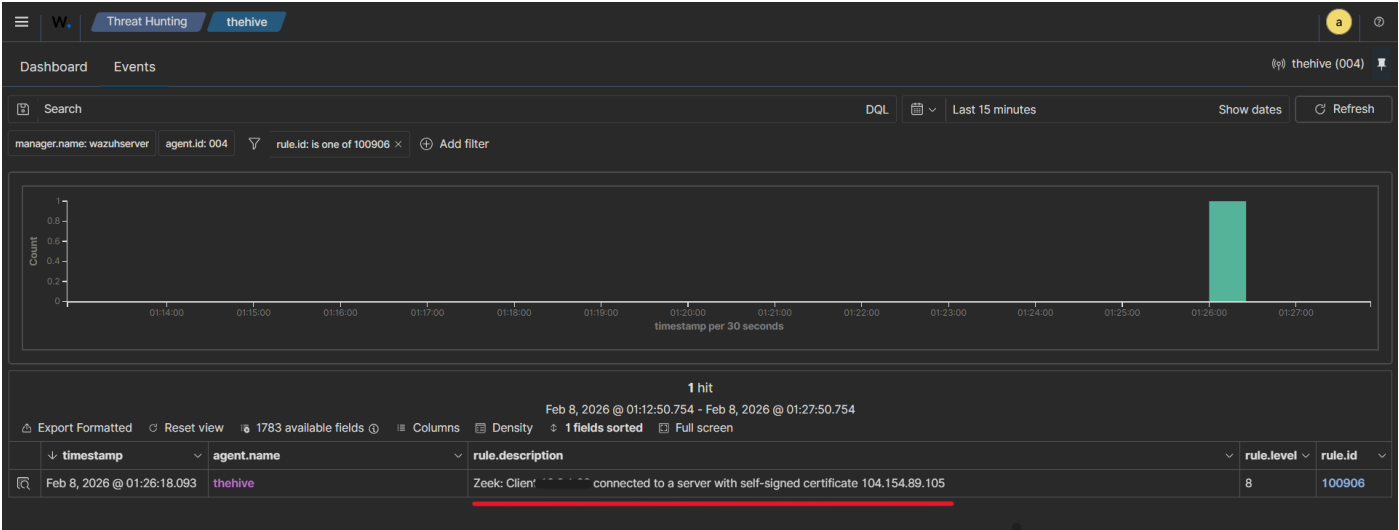
Step 18: SSL/TLS Alerts in Wazuh

Wazuh alerts showed:

- SSL validation failure
- Cipher details
- Destination IP

Purpose:

Highlights encrypted traffic inspection capability.



|   |                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| t | data.dstip                 | 104.154.89.105                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| t | data.dstport               | 443                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| t | data.srcip                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| t | data.srcport               | 52412                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ? | data.ssl_cipher            | TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ? | data.ssl_curve             | secp256r1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ? | data.ssl_established       | false                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ? | data.ssl_history           | sxkni                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ? | data.ssl_next_protocol     | http/1.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| ? | data.ssl_resumed           | false                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ? | data.ssl_validation_status | self signed certificate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| ? | data.ssl_version           | TLSv12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| t | data.uid                   | CztIpq18LU1ZVFCrIg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| t | decoder.name               | json                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| t | full_log                   | {       "ts":1770494171.7742479,       "uid":"CztIpq18LU1ZVFCrIg",       "id.orig_h":,       "id.orig_p":52412,       "id.resp_h":"104.154.89.105",       "id.resp_p":443,       "version":"TLSv12",       "cipher":"TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256",       "curve":"secp256r1",       "resumed":false,       "next_protocol":"http/1.1",       "established":false,       "ssl_history":"sxkni",       "cert_chain_fps":["0f7f9f0b0b190418b70e0a0a353b7a2737bc07b8945dd4b7ba85f17911b3a0a3"],       "client_cert_chain_fps":[],       "validation_status":"self signed certificate"     } |
| t | id                         | 1770494178.841017                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| t | input.type                 | log                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| t | location                   | /opt/zeek/logs/current/ssl.log                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| t | manager.name               | wazuhserver                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| t | rule.description           | Zeek: Client connected to a server with self-signed certificate 104.154.89.105                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| # | rule.firedtimes            | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| t | rule.groups                | zeek                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| t | rule.id                    | 100906                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| # | rule.level                 | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## Step 19: Building SOC Network Monitoring Dashboard

Created a custom **Wazuh dashboard** including:

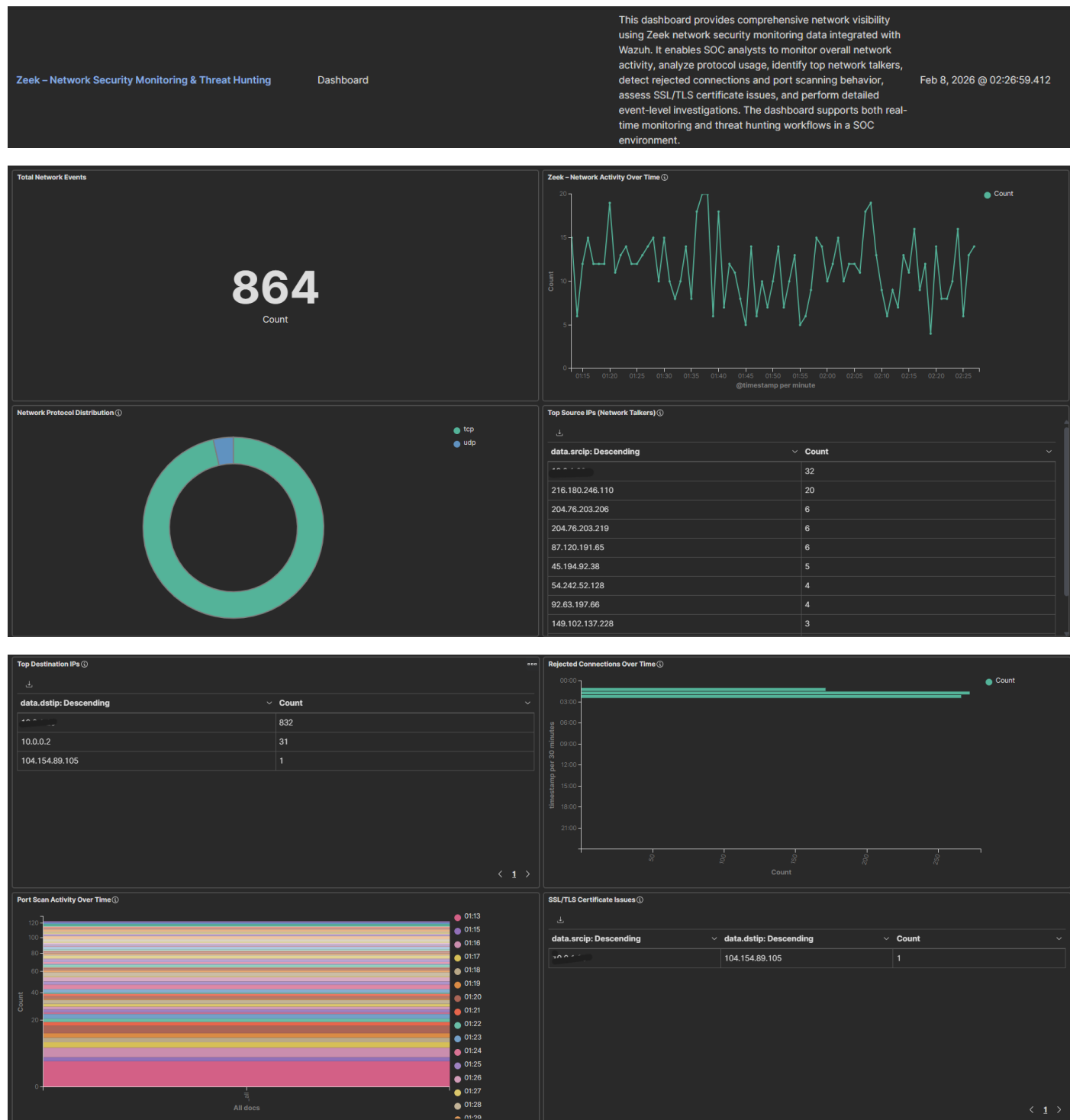
- Total Network Events
- Network Activity Over Time
- Protocol Distribution
- Top Source & Destination IPs
- Rejected Connections Timeline
- Port Scan Activity
- SSL/TLS Certificate Issues

- Raw Network Events Table

## Purpose:

Provides SOC analysts with:

- Fast situational awareness
- Visual threat hunting
- Easier incident investigation



Raw Network Events

| rule.description: Descending                 | data.scrip: Descending | data.dstip: Descending | data.dstport: Descending | data.protocol: Descending | Count |
|----------------------------------------------|------------------------|------------------------|--------------------------|---------------------------|-------|
| Zeek: Connection from 87.120.191.65 to 10.0  | 87.120.191.65          | 10.0.0.0               | 8728                     | tcp                       | 6     |
| Zeek: Connection from 204.76.203.206 to 10.  | 204.76.203.206         | 10.0.0.0               | 17000                    | tcp                       | 4     |
| Zeek: Connection from 193.24.211.239 to 10.  | 193.24.211.239         | 10.0.0.0               | 10002                    | tcp                       | 3     |
| Zeek: Connection from 107.150.96.166 to 10.  | 107.150.96.166         | 10.0.0.0               | 1223                     | tcp                       | 2     |
| Zeek: Connection from 149.102.137.228 to 10. | 149.102.137.228        | 10.0.0.0               | 3128                     | tcp                       | 2     |
| Zeek: Connection from 87.120.191.13 to 10.0  | 87.120.191.13          | 10.0.0.0               | 8728                     | tcp                       | 2     |
| Zeek: Multiple rejected connections from 162 | 162.216.150.185        | 10.0.0.0               | 10444                    | tcp                       | 1     |
| Zeek: Multiple rejected connections from 162 | 162.216.150.185        | 10.0.0.0               | 3000                     | tcp                       | 1     |
| Zeek: Multiple rejected connections from 216 | 216.180.246.110        | 10.0.0.0               | 23231                    | tcp                       | 1     |
| Zeek: Multiple rejected connections from 216 | 216.180.246.110        | 10.0.0.0               | 8280                     | tcp                       | 1     |

## Project Outcome

- Successfully integrated **Zeek with Wazuh**
- Achieved **real-time network visibility**
- Detected DNS abuse, recon activity, and SSL threats
- Built SOC-ready dashboards for analysts
- Enhanced SIEM capabilities beyond host logs

## Skills Gained

- Network traffic analysis
- Zeek configuration & tuning
- SIEM log parsing & correlation
- Custom decoder & rule creation
- Threat detection & simulation
- SOC dashboard design
- Blue team monitoring techniques

## Conclusion

This project showcases the successful integration of **Zeek with Wazuh** to enhance network visibility and threat detection in a SOC environment. By monitoring real network activities and simulating attacks, the setup effectively detected suspicious behavior such as DNS abuse, port scanning, and SSL anomalies. The custom dashboards and alerts enabled efficient analysis and faster incident response. Overall, this integration significantly strengthened network security monitoring capabilities.